An explicit polynomial in twenty-nine variables with effective elliptic sum heights

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Abstract

Under the four arithmetic inputs of the specified elliptic construction, we give an explicitly specified integer polynomial in twenty-nine total variables whose rational input projection is exactly the integers. Its total degree is less than 2^{157} and its coefficient height is at most $2^{2^{168}}$. The construction uses direct square-abscissa charts, a degree-six square predicate, and an effective height bound for the sum of four elliptic abscissas. A single fixed coefficient exponent, 137731, is proved sufficient. There is no ineffective constant in the specification. Every coefficient is determined by integer arithmetic and finite determinant, resultant, and product operations; independent programs verify the source identities, all finite branches, and the degree and coefficient bounds.

1 The defining polynomial and its arithmetic inputs

All variable counts include the distinguished input. A polynomial $F \in \mathbb{Z}[X, Y_1, \dots, Y_m]$ defines $\mathbb{Z}$ over $\mathbb{Q}$ if

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^m \quad F(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

The coefficient height of an integer polynomial is the maximum absolute value of its coefficients. The letter h denotes the absolute logarithmic Weil height, with $h(u/v) = \log \max(|u|, |v|)$ for relatively prime rational integers u, v and $v > 0$.

Theorem 1. *Assume the four arithmetic assertions in Section 1.1. The specification accompanying this paper determines a polynomial*

$$F \in \mathbb{Z}[X, Y_1, \dots, Y_{28}]$$

that defines $\mathbb{Z}$ over $\mathbb{Q}$ and satisfies

$$\deg F < 2^{157}, \qquad \max |\operatorname{coeff}(F)| \le 2^{2^{168}}.$$

Every coefficient choice, finite index range, and arithmetic operation in the specification is explicit. In particular, the fixed odd exponent $L = 137731$ is a coefficient parameter, not a quantified variable.

The determinant specification is a compact exact description of an ordinary polynomial. Matrix entries, finite-algebra generators, and indices of finite products are not additional quantified coordinates. The assumptions below remain explicit: the numerical and algebraic checks do not replace their arithmetic proofs.

1.1 The four inherited inputs

Set

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \quad E : y^2 = x^3 - Dx,$$
$$P = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

The construction uses the following assertions, with exactly the scope established in the arithmetic dependency paper [4].

1. $2E(K) = \mathbb{Z}P$, and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n.

2. For the five normalized contacts with $\infty, 0, 1, 2401, -343/32$, the odd contact radical $M_5(q)$ outside primes at most 76176 satisfies $h(q) \leq HM_5(q)^{19}$ when q is not a contact value, where
$$H = 2^{110}210^{1800}\left(\prod_{p \leq 76176} p\right)^{19} + 128.$$

3. With $B = 17b$, the pole system with preparation parameter zero is
$$p^2 = B(1-e)(1+3e), \qquad Bq^2 = (s^2 - Be(B-b^2))(s^2 + 3Be(B-b^2)).$$
Its solvability excludes negative odd $v_\ell(b)$ for $\ell \notin \{2, 3, 17\}$. It is soluble whenever $0 < b < 17$, its reduced denominator is square, it is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and its valuations there are at least $3, 1, 2$, respectively. We retain the same restricted-converse and descent hypotheses supporting this assertion, and their simultaneous-realization construction. Lemma 3 proves the nonzero-witness consequence of these inputs.

4. The simultaneous prime-pattern input permits the source's prescribed prime values, specified Gaussian generators, and reduction-order conditions, arbitrary fixed divisibility, and arbitrary fixed odd avoidance. It applies to the finite collection of comparison scales after the root points have been fixed.

The prime-pattern input includes its stated application of Sun's theorem. All four assertions have exactly the scope recorded in the arithmetic dependency paper [4]. The effective height lemma used here is proved in the companion paper [2]; it introduces no further arithmetic assumption.

2 Effective control by the two components of a sum

For four nonzero integer labels $n_1, \ldots, n_4$, put

$$Z = \sum_{i=1}^{4} x([n_i]P) = S_0 + S_1\sqrt{2}, \qquad B = \max_i |n_i|.$$

The effective elliptic sum theorem [2] proves

$$B^2 \leq 2h(Z) + 80 \leq 2h(S_0) + 2h(S_1) + 83. \tag{1}$$

Its proof uses formal-group estimates at finite places and positivity at the real embeddings; it applies to all nonzero integer labels, without a parity restriction. The underlying quantitative estimate is

$$h(Z) \geq B^2 - 24\log B - 6 - \log 864.$$

Indeed, $24\log B - B^2/2 < 27$ and $\log 864 < 7$ imply the first inequality in (1); the elementary height bound $h(S_0 + S_1\sqrt{2}) \leq h(S_0) + h(S_1) + \frac{3}{2}\log 2$ implies the second. The companion proof treats cancellations at all finite places and supplies the stated constants.

Lemma 2. *Let H be the contact-height constant in Section 1.1. Set $A_0 = 1$, $A_1 = S_0^2$, and $A_2 = (S_0 + S_1)^2$. Suppose the three comparison contact radicals at $A_r\lambda$ have their kth powers dividing a nonzero integer N. Then*

$$B^2 < 4(H + \log 3)|N|^{19/k} + 82, \qquad B^2 - 6 < 7H|N|^{19/k}. \tag{2}$$

Proof. The abscissas of nonzero doubles are positive at both real embeddings. Thus $S_0 > \sqrt{2}|S_1|$, so $S_0 > 0$ and $S_0 + S_1 > 0$. All three scales are nonzero positive rational squares. The fractional linear transformation $t \mapsto 2 + 1/t$ identifies the five comparison contacts with the five source contacts away from the specified exceptional primes. The contact-height input gives

$$h(A_r\lambda) \le T := (H + \log 3)|N|^{19/k} \qquad (0 \le r \le 2).$$

Taking quotients with the unscaled term gives $h(A_r) \le 2T$, and consequently

$$h(S_0) \le T, \qquad h(S_0 + S_1) \le T.$$

Put $C = S_0 + S_1$ and write

$$Z = (1 - \sqrt{2})S_0 + \sqrt{2}C.$$

The two coefficients are algebraic integers, so the nonarchimedean local contribution is at most the sum of those of S_0 and C. At the two real embeddings the additional factors are bounded by $2\sqrt{2} - 1$ and $2\sqrt{2} + 1$, whose product is 7. Thus

$$h(Z) \le h(S_0) + h(C) + \tfrac{1}{2}\log 7 \le 2T + \tfrac{1}{2}\log 7.$$

Equation (1) now gives $B^2 \le 4T + 80 + \log 7 < 4T + 82$. For the second bound use $|N| \ge 1$, $H \ge 28$, and $4(H + \log 3) + 76 < 7H$. □

3 The exact rational source

We use the point charts, local tests, and denominator guards of the established source [3]. We describe their data and the exact choices used here.

There are nine points over K, named

$$A, S, T_1, Z, H_0, U_A, U_S, U_T, U_Z.$$

For each point retain two rational coordinates h_0, h_1 and set

$$h = h_0 + h_1\sqrt{2}, \qquad x = h^2, \qquad j^2 = h^4 - D, \qquad y = hj. \tag{3}$$

This chart describes precisely the nonzero points of $\mathbb{Z}P$. Indeed, let $R_0 = (1 + \sqrt{2}, 2\sqrt{2})$, so $2R_0 = P$. The group input implies that every $Q \in E(K)$ is $mR_0 + T$ with $T \in E[2](K)$. For the standard two-isogeny descent map, $\delta(R_0) = 1 + \sqrt{2}$ has opposite signs at the real embeddings, whereas the torsion classes are $1, -D, d, -d$, with equal signs. A square abscissa therefore forces m even. Of these torsion classes only 1 is square: $-D, -d$ are negative and $N_{K/\mathbb{Q}}(d) = 7$ is not a rational square. Thus $Q \in \mathbb{Z}P$. Conversely, $\delta(2R) = 1$ gives a square abscissa for every nonzero multiple of P. Neither $h = 0$ nor $j = 0$ is possible in (3). A direct proof of the descent map and all these assertions is given in [5].

The finite root j is represented by one monic quartic, without a finite choice of charts. Write

$$h^4 - D = A + B\sqrt{2}, \quad A = h_0^4 + 12h_0^2h_1^2 + 4h_1^4 - 11, \quad B = 4h_0h_1(h_0^2 + 2h_1^2) + 6.$$

Set

$$c = 2A + B, \qquad d_1 = -A - B, \qquad j' = (c + d_1\sqrt{2})j.$$

Writing $(c + d_1\sqrt{2})^2(A + B\sqrt{2}) = A' + B'\sqrt{2}$ gives

$$A' = 6A^3 - 9AB^2 - 4B^3, \qquad B' = -4A^3 + 6AB^2 + 3B^3.$$

For $j = u + v\sqrt{2}$, direct expansion gives

$$j' = 2(u^3 - 2v^3) + (4v^3 - u^3)\sqrt{2}.$$

Its rational component is nonzero, since 2 is not a rational cube and $j \neq 0$. The multiplier has norm $2A^2 - B^2$, which is nonzero for rational $(A, B) \neq (0, 0)$ because 2 is not a rational square. Thus the reconstruction uses only these two elementary rational obstructions. The general quadratic extension version is proved in [6]. We may therefore use the single chart

$$Y^4 - 4A'Y^2 + 8(B')^2 = 0, \quad Y \neq 0, \qquad j' = Y/2 + (B'/Y)\sqrt{2}, \qquad j = \frac{j'}{c + d_1\sqrt{2}}. \quad (4)$$

The point-chart factors Y, $A'^2 - 2B'^2$, $2A^2 - B^2$, and the quartic derivative are automatically nonzero at every rational retained-base specialization and every algebraic root. Indeed, $A + B\sqrt{2} = h^4 - D$ cannot vanish: taking norms would imply $N(h)^4 = 49$, which has no rational solution. The homogeneous cubic $-4A^3 + 6AB^2 + 3B^3$ has no nontrivial rational zero, because $4T^3 - 6T - 3$ is Eisenstein at 3. Finally,

$$A'^2 - 2B'^2 = (2A^2 - B^2)^2(A^2 - 2B^2)$$

and the discriminant of the quartic in (4) is

$$2^{15}B'^2(2A^2 - B^2)^4(A^2 - 2B^2)^2 \neq 0.$$

The exact square-test proof is given in [7]. These four factors per point may therefore be omitted from the finite-algebra localization product. Every other denominator of the affine point operations remains guarded.

The four root labels (n, a, t, b) are nonzero integers; no parity condition is needed for soundness. The simultaneous completeness construction retains its original odd labels. The scalar shift is now computed directly as $S - P$, in the affine difference chart. Its excluded equal-abscissa cases are $S = \pm P$, both avoided by that construction.

Put $Q_0 = P$, $(Q_1, Q_2, Q_3, Q_4) = (A, S, T_1, Z)$, $U_0 = H_0$, and $V_i = U_i - Q_i$. Write $z = -x/y$ and $f = 1/x$. The truncated formal logarithms are

$$\ell_r(Q) = z(Q) \sum_{j=0}^{r/4-1} c_j (Df(Q)^2)^j, \quad c_0 = 1, \quad c_1 = -2/5, \quad c_j = c_{j-1}\frac{4j - 5}{4j + 1}.$$

Use precisions $(12, 40, 40, 40)$, ratio orders $(10, 39, 39, 39)$, and scalar order 20. With $\beta = 216X^4 + 450459$, the sixteen tests are the ten formal-group tests $f(U_i), \sigma f(V_i)$, the four differences between the paired logarithmic ratios, the direct target test for β, and the direct Lucas polynomial

$$v^2 - u(u - 2)vw + (u - 2)^2w^2 - 1$$

in the three corresponding ratios. Their assigned orders are 1 for each of the first ten tests and $10, 39, 39, 39, 10, 39$ for the remaining six. These tests and their construction are unchanged.

3.1 Three comparison scales

We use precisely the three positive rational square scales

$$A_0 = 1, \qquad A_1 = S_0^2, \qquad A_2 = (S_0 + S_1)^2, \qquad S_0 + S_1\sqrt{2} = \sum_{i=1}^{4} x(Q_i). \tag{5}$$

They are nonzero: the root points are nonzero doubles and their abscissas are positive in both real embeddings. In fact $x(Q_i) > 3 - \sqrt{2}$ at each embedding, by the curve equation and exclusion of two-torsion. Taking the sum and its half-trace gives $S_0 > 12$, so $A_1 \neq A_0$. For the retained slope λ, each scale is inserted into the same normalized comparison detector of [3]:

$$t_r = A_r\lambda, \quad \delta_r = t_r - 1, \quad F_r = \prod_{\theta\in\Theta}(t_r - \theta), \quad R_r = \frac{F_r}{A_r^5\delta_r^6},$$
$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\}.$$

$$W_r = \prod_{j=1}^{16}\left(1 + R_r\left(\frac{D_j}{\delta_r^{m_j+1}R_r^{m_j}}\right)^2\right), \qquad b_r = \frac{R_r}{\mathrm{Tr}(W_r^{-1})^2}.$$

All displayed denominators and the inherited guards are retained, except for the automatic point-chart factors proved above. Each b_r receives a pole certificate.

For each pole put $B = 17b$. We use the single rational chart

$$e = \frac{\tau^2 - B}{\tau^2 + 3B}, \qquad p = \frac{-4B\tau}{\tau^2 + 3B}, \qquad \tau^2 + 3B \neq 0,$$

with retained parameters τ, s, and require the remaining quadratic pole root q to be nonzero. The following lemma proves completeness of this restriction under the inherited arithmetic inputs.

Lemma 3. *Suppose that b has square reduced denominator,*

$$0 < b < 17, \qquad b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \qquad v_2(b) \geq 3, \quad v_3(b) \geq 1, \quad v_{17}(b) \geq 2.$$

The inherited pole-realization proof supplies a certificate with $pqs \neq 0$ and $e \neq 1$. Consequently the single chart above, with $q \neq 0$, suffices. The restricted predicate has the same pole obstruction, and adds no arithmetic hypothesis.

Proof. Set $V = B(B - b^2)$. Its positivity and its local square conditions are precisely those verified in the zero-preparation argument of [4]. Let β_0, θ be the positive squarefree representatives of B, V. Their supports are disjoint, $\beta_0 \equiv 17 \pmod{24}$, $\theta \equiv 1 \pmod{24}$, and both mutual local-square conditions hold.

The inherited simultaneous-realization proof in [8], with source identifier

`lem:original-seven-simultaneous-realization`,

constructs points on $E_n : y^2 = x(x - n)(x + 3n)$ and $E_{n\theta}$ with nonzero abscissas of squareclass β_0. Both are outside two-torsion. Indeed, the construction includes a fresh prime of n outside $\beta_0\theta$, so β_0 differs in squareclass from n and $n\theta$; its positive squareclass also differs from their negative multiples by -3. These are the only nonzero two-torsion abscissas. This non-torsion assertion is also explicit in the source descent proof. Thus both ordinates are nonzero.

Writing $V = \theta c^2$ and rescaling as in that proof gives

$$(B, y_1) \in E_{e_0}(\mathbb{Q}), \qquad (Bs^2, y_2) \in E_{e_0V}(\mathbb{Q}), \qquad sy_1y_2 \neq 0.$$

The substitutions $e = e_0/B$, $p = y_1/B$, $q = y_2/(B^2 s)$ give exactly the two pole equations and $pqs \neq 0$. The first conic excludes $e = 1$. Set $\tau = p/(e-1)$. Its equation gives

$$\tau^2 + 3B = \frac{4B}{1-e} \neq 0,$$

and then gives the displayed reconstruction of e, p. Every restricted witness is an original pole witness, so its obstruction is unchanged. The construction establishes completeness for every input satisfying the displayed conditions. No equivalence of the two pole predicates for arbitrary rational b is asserted. □

3.2 A linear scalar normalization

The fixed scalar normalization is

$$\begin{gathered} \ell = 76249, \quad r = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1), \\ R_L = \frac{102^2 \ell^L r}{(1+C_L r)^2}, \quad C_L = 17(\ell^{(L+1)/2}+1). \end{gathered} \tag{6}$$

Here $L = 137731$ is fixed, and thus $(L+1)/2 = 68866$. The scalar norm tree and its six rational component tests are unchanged: they concern $f(H_0), f(S-P), z(S-P)/z(H_0)$ with orders $1, 1, 1, 1, 20, 20$. Its output W supplies the scalar pole input $R_L W^2$.

The finite residue certificate verifies that the five contact values are not attained by $216x^4 + 450459$ modulo ℓ, and that $1 + 17r$ never vanishes there. Consequently, for every rational input,

$$v_\ell(R_L) = \begin{cases} L, & v_\ell(X) \geq 0, \\ L - 20v_\ell(X), & v_\ell(X) < 0. \end{cases} \tag{7}$$

These valuations are positive and odd. At a consumed contact prime different from ℓ, the denominator is a unit and the contact valuation is unchanged. For integral X, the reduced denominator of R_L is square, it has the required local squareclasses and depths at $2, 3, 17$, and

$$0 < R_L < \frac{36}{\ell r} < 1.$$

The scalar norm argument therefore gives

$$v_\ell(a-1) \geq 19L + 1. \tag{8}$$

For completeness, if $v_\ell(X) < 0$, the leading coefficient of r is an ℓ-unit, so $v_\ell(r) = 20v_\ell(X) < 0$ and $v_\ell(1 + C_L r) = v_\ell(r)$, proving (7). If $v_\ell(X) \geq 0$, finite-field avoidance makes both r and $1 + C_L r$ units. At a consumed finite contact prime $p \neq \ell$, one has $v_p(r) > 0$, so $1 + C_L r$ is a unit. The real bound follows from $(1 + C_L r)^2 > C_L^2 r^2 > 17^2 \ell^{L+1} r^2$. For integer input, the denominator is an integer square; ℓ and 102^2 preserve the three required local squareclasses, and the normalizer is a unit at $2, 3, 17$. The inherited scalar norm argument applied at the forced contact ℓ gives (8). The pole preparation and simultaneous realization are the same as in [3]. The supplied checker independently verifies every residue modulo 76249 and the normalization identity as an exact polynomial identity.

4 Soundness with the fixed exponent

Let a rational source solution be given. The point charts yield nonzero integer labels (n, a, t, b) for the four root points. The affine guard for $S - P$ excludes $a = 1$. Put

$$c = |a-1|, \qquad B = \max(|n|, |a|, |t|, |b|), \qquad \Delta = t^2 - a(a-2)tb + (a-2)^2 b^2 - 1.$$

The fixed-prime depth gives

$$c \geq \ell^{2616890}, \qquad 16 \cdot 2616890 = 41870240 > 41869080 = 20(2093448 + 6).$$

Since $\ell > 2^{16}$ and $H < 2^{2093448}$, it follows that

$$c > (64H)^{20}. \tag{9}$$

In particular $B \geq c - 1 > 4$, and $|\Delta| \leq 5B^4$. The unchanged formal-logarithm and scalar-norm arguments of [3] give

$$h(\beta) \leq Hc^{19/20}, \tag{10}$$

$$\Delta \neq 0 \Longrightarrow B^2 - 6 < 7H(5B^4)^{19/39}, \tag{11}$$

$$\beta \neq n \Longrightarrow B^2 - 6 < 7H(2e^{h(\beta)}B)^{19/10}. \tag{12}$$

For the last two implications, the local tests give the required integer divisibilities and Lemma 2 supplies their height consequence. The scalar inequality is unchanged because the contact valuations consumed outside ℓ are unchanged by the linear normalizer.

Equation (9) implies $h(\beta) < c/64$. If $\Delta \neq 0$, use $5^{19/39} < \sqrt{5} < 16/7$ and $B^2 > 12$ in (11) to obtain

$$B^{2/39} < 32H.$$

On the other hand,

$$B \geq c - 1 > c/2 > (64H)^{20}/2 > (32H)^{39/2},$$

a contradiction. Thus $\Delta = 0$. The elementary Lucas growth lemma of [3] applies to these nonzero integer labels and gives $\log B > c - 4$. Its hypotheses require $|a| > 7$, which follows from (9).

If $\beta \neq n$, equation (12) gives

$$\log B < 10 \log(64H) + 19h(\beta) < 29c/64 < c - 4.$$

Here $\log(64H) < c/64$ follows from $H \geq 1$ and (9). This contradicts the Lucas growth bound. Consequently $\beta = n \in \mathbb{Z}$. If $X = u/v$ is in lowest terms with $v > 0$, integrality of $216X^4 + 450459$ implies $v^4 \mid 216 = 2^3 3^3$. Hence $v = 1$ and $X \in \mathbb{Z}$. This proves soundness.

5 Completeness

Fix an integer input X and the stated fixed exponent L. All numerator and denominator primes of (6) form a finite set. The source realization first chooses the odd root labels with the scalar label $a - 1$ sufficiently divisible and deep at these primes, and with the corresponding points in the required real neighborhoods. The explicit Lucas construction then gives odd nonzero t, b with $\Delta = 0$ and arbitrarily large growth. The input label is chosen to be $n = \beta$.

At this stage the three scales in (5) are fixed positive rational squares. Remove repetitions before applying the simultaneous prime-pattern theorem. At least two distinct scales remain, since $S_0 > 12$; thus its hypothesis requiring at least two scales is satisfied. Equal scales use the same detector and pole witnesses in the source. The theorem chooses the slope and contact primes with the prescribed conjugate reduction orders while avoiding the previously fixed odd modulus. The subsequent Chinese remainder choice of the auxiliary index h and the points $U_i = [hn_i]P$ supplies the sixteen local tests to their assigned precisions and preserves the fixed scalar requirements and real neighborhoods. This is precisely the simultaneous construction of [4, 3]; its prime-pattern input accepts any fixed finite collection of the positive square scales and any fixed finite avoidance modulus.

Each of the three comparison inputs is thereby prepared for the zero-preparation pole theorem. The scalar input is prepared as well: R_L has square reduced denominator and the required local squareclasses, the six normalized scalar tests can be made arbitrarily small at the fixed finite places, and the scalar norm output can be made arbitrarily close to 1 in the real absolute value. Thus $0 < R_L W^2 < 17$, with the requisite denominator and local conditions. The four pole certificates exist by the inherited completeness input. The direct point chart (4) covers all nine chosen points; its multiplier needs no further choice. Lemma 3 applies independently to each pole input. Its parameters belong only to that pole, so the certificates can be chosen simultaneously without any common-parameter assumption. All pole roots are nonzero and the single conic chart applies. All remaining inherited guards are maintained by the same realization.

This proves completeness for the fixed specification. Together with the preceding section it proves that the rational input projection of the source is exactly $\mathbb{Z}$.

6 Finite-algebra elimination and the exact bounds

The retained rational coordinates are

Coordinates	Number
Input	1
Nine square-root-abscissa pairs	18
Comparison slope	1
Three comparison-pole parameter pairs	6
Scalar-pole parameter pair	2
Total retained before elimination	28

There are nine monic quartic point roots and four monic quadratic pole roots. The first conics use one chart, and every quadratic pole root is already guarded nonzero in the source. Thus zero-root strata are inconsistent and are discarded. The finite-algebra rank is $4^9 2^4 = 2^{22}$. The point quartics are separable at every rational retained base. The direct-scaled finite-algebra theorem of [1] replaces all thirteen finite roots by one rational coordinate. There are precisely twenty-nine total polynomial coordinates.

Let n_0 be the rank of the finite algebra and let m be its number of monic generators $z_1, \dots, z_m$. Let G be the product of the remaining nonzero guards. For each integer

$$0 \le s \le m n_0 (n_0 + 1)/2, \qquad \lambda_s = 1 + \sum_{i=1}^{m} s^i z_i,$$

form the multiplication matrix and its characteristic polynomial

$$H(V) = \det(I - V M_{G\lambda_s}) = \sum_{k=0}^{n_0} a_k V^k.$$

There are no closed residual equations in this source. For every $1 \le d \le n_0$, define

$$H_d(V) = \sum_{k=0}^{d} a_k V^k, \qquad g_d = \mathrm{Res}_V(H_d, H_d'),$$

using Sylvester degrees $d, d-1$. The factor for this case is

$$Q_{s,d}(U) = \begin{cases} H_d(g_d U)^2 + \displaystyle\sum_{k=d+1}^{n_0} a_k^2, & d < n_0, \\ H_{n_0}(g_{n_0} U), & d = n_0. \end{cases}$$

The defining polynomial is the product of these factors over all specified s, d. Every operation is integer coefficient arithmetic with explicit finite index ranges.

The direct-scaled finite-algebra theorem of [1] proves that this product has exactly the rational base projection of the guarded source. Indeed, $H_d(0) = 1$ excludes $g_d = 0$ at every zero of a factor, and the resultant then makes the full characteristic polynomial squarefree of degree d. Its rational root gives a one-dimensional rational eigenspace for all commuting multiplication matrices, hence a guarded rational point. Conversely one of the specified separating elements makes all active eigenvalues distinct and nonzero. Inactive nonreduced fibers, complex residue factors, and drops in degree are included in that proof.

For the quantitative bounds, give each retained variable weight one and each monic generator its certified weight. If the multiplication element $G\lambda_s$ has weight at most w, then $\deg a_k \le kw$: the basis-weight differences cancel in every principal minor. The resultant has coefficient index weight d^2. Consequently, with

$$S = 1 + \frac{mn_0(n_0+1)}{2}, \qquad C(n_0) = 2\sum_{d=1}^{n_0-1} \max(d^3+d, n_0) + (n_0^3+n_0),$$

we have

$$\deg F \le S\big(wC(n_0) + n_0^2\big).$$

The complete degree and coefficient-norm proofs are supplied in [1].

Here $m = 13$, $n_0 = 2^{22}$, and the generator weights are

$$\underbrace{6, \dots, 6}_{9 \text{ entries}}, 1568138,\ 1935231,\ 1935231,\ 116318.$$

The guard weight is 6789625, and hence $w = 8724856$. Weighted companion-column norms give the uniform logarithmic operator bound $\beta = 12887537402$. There are 479615460265661825024 factors. The integer degree majorant is

$$D_* = 154383598645603405051555717167316362269845989152 < 2^{157}.$$

The coefficient-norm theorem of [1], with $r = 22$, gives

$$\log_2 \|F\|_1 \le S\big((\beta+r)C(n_0) + (r+1)(2n_0^3 + 4n_0 - 2)\big) = C_*,$$

where

$$C_* = 228040944767112830511621792043639838233373391751570 < 2^{168}.$$

The coefficient height is at most the sum of absolute coefficients $\|F\|_1$. These are upper bounds; no assertion that the expanded polynomial attains either bound is needed. All finite sums and integer comparisons are independently checked. This proves the bounds in Theorem 1.

7 Reproduction and computational scope

The directory `effective29_nontorsion` contains the exact compressed source and all programs for its construction and independent source verification. From that directory,

```
python3 reproduce.py
```

reconstructs the source at the proved sufficient exponent 137731. Its source has 7944 coefficient nodes, and its integer coefficient graph has 3243 nodes. Checks cover nine quartic point charts, four rational-square clearings, all 114 atomic source guard factors, and the 36 automatic factors omitted from the guard. The scalar normalizer is checked at all 76249 residue classes, and its cutoff inequalities are checked by exact integer comparisons.

The directory `weighted_elimination` contains the compiler and independent checker:

```
build_weighted_specification.py
check_weighted_specification.py
```

The final specification is in `weighted29`. The compiler uses the verified source coefficient graph and writes precisely the polynomial defined above. The checker evaluates the degree and coefficient-bound sums literally, and checks small exact finite algebras with complex residue factors, inactive nilpotents, zero guards, selector collisions, and degree drops. All programs require only the Python standard library.

These computations verify the specified algebra and numerical bounds. The nonvanishing arguments, effective height theorem, and four arithmetic inputs have the proofs and scope stated above. Exact finite coefficient reconstruction does not prove those global arithmetic premises.

References

[1] *Weighted bounds for eliminating finite algebraic choices*, companion paper, 9 November 2026, `weighted_finite_algebra_20261109_v12.tex`.

[2] *Effective heights of sums of elliptic abscissas*, companion paper, 8 November 2026, `effective_elliptic_sums_20261108_v10.tex`.

[3] *An explicit polynomial in 29 variables over the rationals*, the elliptic source, Lucas growth lemma, and guarded finite-algebra construction, version of 8 November 2026, `few_variables_20261108_v8.tex`. https://github.com/JoseBrox/diophantine-z-q_improvements.

[4] *Arithmetic dependencies of the construction with 29 variables*, version of 8 November 2026, `arithmetic_dependencies_20261108_v8.tex`. https://github.com/JoseBrox/diophantine-z-q_improvements.

[5] *Square abscissas and direct charts for the elliptic subgroup*, companion paper, 8 November 2026, `square_abscissa_charts_20261108_v9.tex`.

[6] *A single quartic chart for square roots in a quadratic extension*, companion paper, 8 November 2026, `quadratic_square_chart_20261108_v9.tex`.

[7] *Optimal polynomial square tests in quadratic number fields*, companion paper, 8 November 2026, `quadratic_square_predicates_20261108_v10.tex`.

[8] E. Glazer, *Diophantine definition of the integers over the rationals*, Version 9, `manuscripts/quartic_v9.tex`, commit `14e8f601fe0d9173d4a5f084aedc3e07066a7688`. https://github.com/elliotglazer/diophantine-z-q. The inherited restricted-converse correction is identified in [4].